WattWise Malaysia: Official Project Proposal

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1. Executive Summary

WattWise Malaysia is an AI energy advisor that turns a TNB electricity bill into one clear, low-effort action plan. This proposal seeks approval and funding for a six-month pilot.

Households and small businesses receive a bill they cannot easily interpret and cannot afford to have audited. WattWise reads the kWh and RM figures a client shares, estimates what is driving the cost, and ranks practical actions by impact and effort. Every reply ends with one next step that takes under five minutes.

- **Who it serves:** Malaysian households and small businesses on standard TNB tariffs.
- **What it delivers:** a consumption summary, main energy drivers, ranked actions, a monthly monitoring routine and one next step.
- **Pilot scope (proposed):** six months, 500 households and 50 small businesses.
- **Funding requested:** RM30,800, indicative and subject to quotation (see Budget).
- **Success test:** a measured reduction in kWh among active users. Savings are estimated and never guaranteed.

2. Problem Statement

Most Malaysian electricity customers learn what they used only after the month has ended, in a bill they find hard to read. That gap between usage and understanding is where avoidable cost builds up.

- **Time-poor clients.** Households and small-business owners review the bill, at best, once a month and rarely look past the total in RM.
- **Confusing structure.** Tariff blocks (price bands that change with how much you use) and bill add-ons make it hard to see why a bill rose.
- **Costly expertise.** Professional energy consultations are priced for large facilities, not for a family home or a corner café.
- **Generic advice.** Common energy-saving tips are not tied to a client's own kWh, appliances or schedule, so they are easy to ignore.

- **Tropical load.** Air conditioning, refrigeration and long cooling hours in a hot, humid climate make cooling habits the most likely lever for many clients.

The opportunity is a service that is fast, affordable and specific to the client's own numbers.

3. Proposed Solution

WattWise is a conversational advisor that analyses a TNB bill and replies in a fixed, scannable five-part format, in Bahasa Malaysia, English or a mix.

1. **Consumption Summary.** Total kWh and RM billed, with the month-to-month change in kWh and %.
2. **Main Energy Drivers.** Estimated contribution of air conditioning, water heating, refrigeration, lighting and standby power, using kWh = kW × hours. Bill data and estimates are labelled separately.
3. **Recommended Actions.** Three to four actions ranked by impact and effort: zero-cost habits first, then low-cost fixes, then larger purchases only when payback is clear. Each shows an estimated saving as a range, its assumptions and any comfort trade-off.
4. **Simple Monitoring Routine.** Once a month, track three figures: kWh, RM and % change.
5. **Next Step.** One action that takes under five minutes.

Worked example. A household of four uses 650 kWh (RM250) against 520 kWh the month before, a rise of 130 kWh (25%). With two air conditioners running about eight hours a day, WattWise flags cooling as the likely driver. It suggests a 24–25°C setting, sleep timers and filter cleaning, and sets a target of about 550 kWh. The next step is to set the sleep timers tonight.

Design principles

- Useful advice even when data is incomplete: assumptions are stated and at most one follow-up question is asked.
- Ranges instead of false precision, and no guaranteed savings.
- Set-and-forget actions (timers, smart plugs, scheduled use) over daily manual tracking.
- Tariff rates come from a maintained reference table, not from the model's memory.

Boundaries. WattWise does not give legal, tariff-contract or safety-critical electrical advice. For faults, unsafe wiring or burning smells, it tells the client to contact a qualified electrician or TNB.

4. Target Market and Value Proposition

WattWise starts with time-poor households and micro-businesses, where bills are large enough to matter and consultancy is out of reach.

Segment	Pain point	WattWise value
Urban households	Bill rises with no clear cause	Plain-language explanation and a monthly kWh target
Families with long cooling hours	Air conditioning dominates usage	Timer and temperature actions with kWh and RM ranges
Renters and students	No control over appliances or wiring	Zero-cost habit changes only
Small businesses (cafés, kiosks, retail, home-based)	Refrigeration and cooling run all day	Appliance-level estimates and low-disruption fixes

Value proposition: professional-style bill analysis in minutes, at a fraction of consultancy cost, with one concrete action each time.

5. Business Model

WattWise runs a freemium model: a free basic analysis builds trust and reach, while paid plans and institutional partners fund the service. The one-page infographic Business Model Canvas that accompanies this proposal shows the same nine blocks visually. Prices below are hypotheses to be tested in the pilot.

Block	Summary
Customer segments	Households; small businesses; later, property managers and housing communities
Value propositions	Fast bill analysis, ranked actions, one next step, bilingual, low effort
Channels	Web chat and messaging apps; social media; community, school and university networks; partner referrals

Block	Summary
Customer relationships	Self-service chat; monthly bill-check reminder; community education content
Revenue streams	Free tier; premium plan for households and SMEs (indicative RM5–15 per month, to be tested); licensing to partners; sponsored or grant-funded campaigns
Key resources	Advice engine and tested prompt; maintained TNB tariff table; calculation checks; brand and trust
Key activities	Product testing; tariff data upkeep; accuracy audits; content and outreach; partnerships
Key partners	Universities and youth bodies; energy-efficiency and sustainability organisations; appliance retailers; banks and property managers; relevant government agencies
Cost structure	AI usage and hosting; team and testing; marketing and content; compliance and privacy review; data upkeep

6. Impact and Alignment

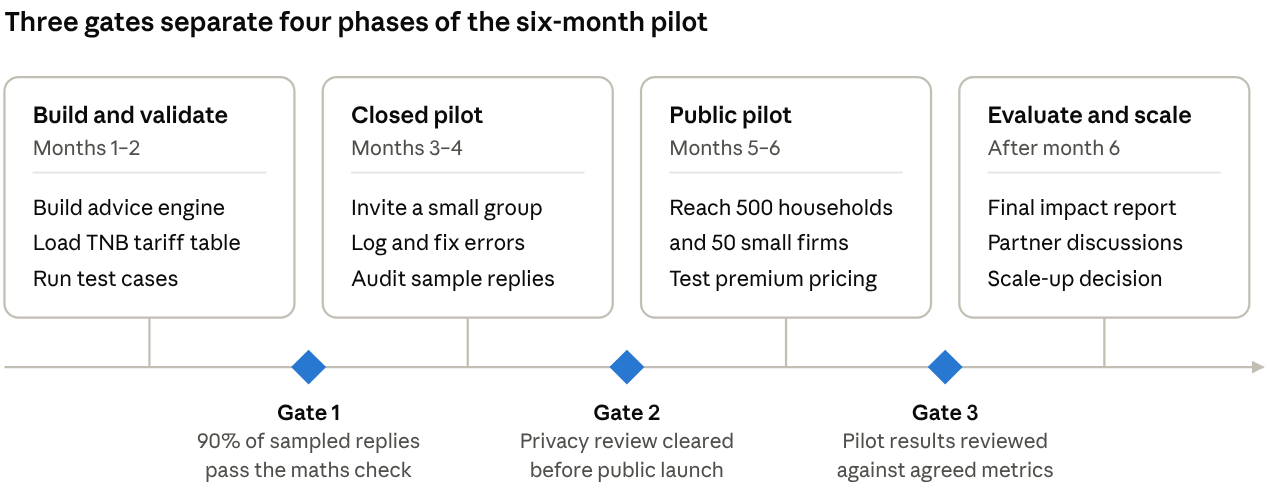
The aim is lower bills for clients and less electricity demand for the grid, achieved through many small, repeatable changes.

- **Household impact.** Lower kWh and RM for clients who act on one or two recommendations, measured against their own previous months.
- **Energy literacy.** Clients learn what a kWh is, how tariff blocks work and which appliances matter most.
- **Accessibility.** A free tier and plain language open energy advice to people who would never book a consultant.
- **National and global goals.** The service supports energy conservation and Malaysia's wider energy-transition agenda, and aligns with UN SDG 7 (affordable and clean energy), SDG 12 (responsible consumption) and SDG 13 (climate action).

Impact will be reported as measured change among active users, with the method and limits stated openly.

7. Implementation Plan and Timeline

The pilot runs in four phases with three review gates, so errors are caught while the user group is still small.



pilot roadmap · 4 phases, 3 gates

Each gate must be passed before the next phase starts. Phases are equal bands and not drawn to scale.

8. Budget and Resource Requirements

The six-month pilot needs RM30,800, including a 10% contingency. All figures are planning estimates to be replaced by quotations before approval.

Item	Scope	Indicative cost (RM)
Product build and testing	Advice engine, prompt refinement, calculation checks, test cases	5,000
AI usage and hosting	Six months of pilot traffic	6,000
Pilot recruitment and user testing	500 households, 50 small businesses	6,000
Marketing and content	Social content, community outreach, printed and digital materials	5,000

Item	Scope	Indicative cost (RM)
Privacy and compliance review	Data handling and consent wording	3,000
Evaluation and reporting	Baseline, follow-up, final report	3,000
Subtotal		28,000
Contingency (10%)		2,800
Total request		**30,800**

Non-cash resources needed: a maintained table of current TNB domestic and small-business rates, access to pilot communities, and a named reviewer for energy-advice accuracy.

9. Risks, Ethics and Safeguards

The main risk is confident but wrong advice, so accuracy and honest limits are built into the service.

Risk	Safeguard
Inaccurate appliance estimates	Show ranges, label estimates versus bill data, flag when appliance totals do not match the billed kWh
Outdated tariff information	Maintained reference table with an "as of" date; no rates quoted from model memory
Over-promised savings	Wording such as "could save roughly"; no guarantees; assumptions stated
Unsafe electrical advice	Out of scope; refer faults and wiring concerns to a qualified electrician or TNB
Personal data and privacy	Collect only what is needed, obtain consent, and follow the Personal Data Protection Act 2010; review before launch
Low adoption	Free tier, bilingual replies, and distribution through trusted community and campus networks

Risk	Safeguard
AI cost overrun	Usage caps, monitoring and a contingency line in the budget

10. Success Metrics and Monitoring

The pilot succeeds if clients act on the advice and their usage falls, with accuracy held to a checkable standard. Targets are proposals to be confirmed with the approving body.

Metric	Proposed target	How measured
Pilot users reached	500 households, 50 small businesses	Registrations
Clients who adopt at least one action	40% or more	Short monthly check-in
Change in monthly kWh among active users	5–10% median reduction	Bill kWh against each client's own earlier months
Client satisfaction	4 out of 5 or higher	In-chat rating
Responses passing the arithmetic check	90% or more	Monthly audit of a random sample

Progress is reviewed monthly on three figures: users, adoption rate and median kWh change. Results, including any shortfall, go into the final pilot report.

11. Conclusion and Call to Action

WattWise gives time-poor Malaysians what a consultant would, in minutes and at low cost: a clear reading of the bill and one action to take today. We request approval of the six-month pilot and the RM30,800 budget.

Before submission, please confirm:

- [ ] Final project name and proposing organisation
- [ ] Pilot scope, targets and budget
- [ ] Current TNB tariff table to load into the advice engine

- [ ] Pilot partners and the named accuracy reviewer